Examiner only

8. The diagram below shows a hydraulic jack system.
The pressure in the fluid throughout the system is constant.

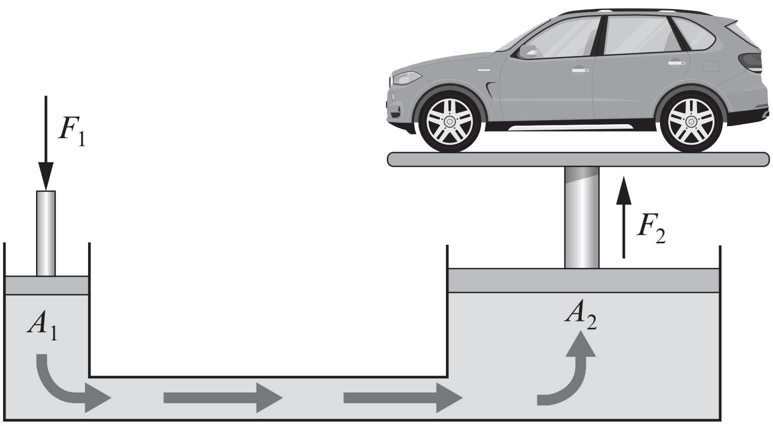

Diagram not drawn to scale

(a) A force F_1 of size 25 N is applied to a piston of area A_1 equal to 10 cm^2.
Use an equation from page 2 to calculate the pressure created in the fluid. [2]

pressure = ... N/cm^2

(b) Use the equation:

force = pressure × area

to calculate the force F_2 if the area A_2 equals 4000 cm^2. [2]

force = ... N

(c) The jack works because hydraulic fluids cannot be compressed.
Explain, **in terms of molecules**, why the jack will not work if it contains air. [2]

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